Database Lab : Tablespace Capacity Utilization & Autoextend Audit

Duration : 30 Min.

Scenario

As a Database Capacity Specialist at NIIT, unexpected halts occur when tablespace datafiles reach capacity without autoextend. You must establish a capacity audit table in PostgreSQL, calculate utilized percentages, and flag datafiles requiring expansion.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **PostgreSQL Service:** `localhost:5432` | DB: `postgres` | User: `postgres` | Password: `PostgresPassword123!` | CLI: `psql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **Capacity Audit Table:** `<prefix>_tablespace_capacity`

Task Objectives

Perform the following configuration steps directly via `psql`:

1. Create Tablespace Capacity Audit Table

- Create table `<prefix>_tablespace_capacity` with columns:
- `tablespace_name VARCHAR(50) PRIMARY KEY`
- `allocated_mb INT NOT NULL`
- `used_mb INT NOT NULL`
- `used_pct NUMERIC(5,2) NOT NULL`
- `autoextend_enabled BOOLEAN NOT NULL`
- `risk_level VARCHAR(20) NOT NULL`

2. Insert Tablespace Risk Metrics

- Insert the following 3 capacity assessments:
- `('USERS_TBS', 5000, 2100, 42.00, TRUE, 'LOW')`
- `('DATA_TBS', 10000, 9850, 98.50, FALSE, 'CRITICAL')`
- `('INDX_TBS', 8000, 4200, 52.50, TRUE, 'MEDIUM')`

3. Verify Risk Ranking Query

- Query the critical tablespaces: `SELECT * FROM <prefix>_tablespace_capacity WHERE risk_level = 'CRITICAL';`.

Verification

Once completed, run the verification script:

```
python run.py
```

The driver connects to PostgreSQL and validates the capacity audit table.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Capacity Table (_tablespace_capacity) Creation	3 Marks
TC2	Utilization Metric & Percentage Calculation Population	3 Marks
TC3	Autoextend Off Critical Risk Level Flagging	2 Marks
TC4	Capacity Risk Ranking & Query Integrity	2 Marks

Total Score: 10 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- Ensure used_pct matches the allocated and used values.